Covering Factor Definitions in the LaRT Clumpy Medium

Contents

1. Setup and Notation

We consider a spherical domain of radius R filled with a population of spherical clumps. In the present LaRT implementation each clump has the same radius r_c and the same internal opacity, but the formulas below are written so that a generalization to position-dependent $r_c(\mathbf{r})$ and clump number density $n_c(\mathbf{r})$ is immediate.

The fundamental local quantity is the differential covering rate per unit path length,

$$\kappa_{\rm cov}(\mathbf{r}) \;\equiv\; n_c(\mathbf{r})\,\pi\, r_c(\mathbf{r})^2, \tag{1}$$

which has units of inverse length and represents the expected number of clumps a sightline encounters per unit length when passing through the neighborhood of $\mathbf{r}$.

For uniform profiles $n_c(\mathbf{r}) = n_c^{(0)} \equiv N/[(4/3)\pi R^3]$ and $r_c(\mathbf{r}) = r_c$, so $\kappa_{\rm cov} = N r_c^2/((4/3)R^3)$.

2. Current LaRT Definition (Uniform Clumps)

LaRT currently uses

$$\boxed{f_{\rm cov}^{\rm LaRT} \;=\; \tfrac{3}{4}N\left(\frac{r_c}{R}\right)^2} \tag{2}$$

as the covering factor associated with N clumps of radius r_c packed into a sphere of radius R (see `clump_mod.f90`, lines 93 and 136). The companion volume filling factor is

$$f_{\rm vol}^{\rm LaRT} \;=\; N\left(\frac{r_c}{R}\right)^3. \tag{3}$$

Equation (2) is exactly the radial line-of-sight integral of $\kappa_{\rm cov}$ from the box center to the surface,

$$f_{\rm cov}^{\rm LaRT} \;=\; \int_0^R \kappa_{\rm cov}\, dr \;=\; n_c^{(0)}\,\pi\, r_c^2\, R \;=\; \tfrac{3}{4}\, N\, (r_c/R)^2, \tag{4}$$

i.e. the average number of clumps a photon emitted at the center crosses on its way out. This is the natural Monte-Carlo interpretation: it equals the expected count of clump intersections along a single radial trajectory emerging from the source.

3. Generalized Definitions for Non-Uniform Profiles

When r_c and n_c depend on position, the single number $f_{\rm cov}$ must be defined by a particular averaging prescription. Three natural choices are described below. All three reduce to Eq. (2) (up to a constant prefactor) in the uniform limit.

3.1 Radial Line-of-Sight Integral (current LaRT convention)

Integrate $\kappa_{\rm cov}$ along a single radial sightline from the box center to the surface,

$$f_{\rm cov}^{\rm LOS} \;=\; \int_0^R n_c(r)\, \pi\, r_c(r)^2\, dr. \tag{5}$$

For a spherically-symmetric profile the result is direction-independent.

Uniform limit. Substituting $n_c = n_c^{(0)}$ and constant r_c recovers exactly Eq. (2).

Physical meaning. The expected number of clumps a photon launched at the center traverses on its way to the surface. This is the quantity that controls the radiative-transfer outcome for a centrally located source most directly.

3.2 Impact-Parameter Average (Random External Sightline)

A distant external observer sees the box covered by chords of varying impact parameter $b \in [0, R]$. Averaging the covering along such chords with the natural area weight yields

$$\langle f_{\rm cov} \rangle_b \;=\; \frac{1}{\pi R^2} \int_0^R 2\pi b\, f_{\rm cov}(b)\, db, \tag{6}$$

where $f_{\rm cov}(b)$ is the line integral of $\kappa_{\rm cov}$ along the chord at impact parameter b.

Uniform limit. The chord length is $2\sqrt{R^2 - b^2}$, so

$$\begin{aligned}\langle f_{\rm cov} \rangle_b^{\rm uniform} &= n_c^{(0)}\,\pi\, r_c^2\, \frac{4}{R^2} \int_0^R b\sqrt{R^2 - b^2}\, db \\ &= \tfrac{4}{3}\, n_c^{(0)}\,\pi\, r_c^2\, R \;=\; N \left(\frac{r_c}{R}\right)^2,\end{aligned} \tag{7}$$

which exceeds Eq. (2) by a factor $4/3$.

Physical meaning. The covering factor as inferred from absorption observations of a background source for which all sightlines through the cloud are equally likely (uniform illumination by a distant continuum source).

3.3 Volume-Weighted Average

Volume-average the local covering rate $\kappa_{\rm cov}(\mathbf{r})$ and multiply by the radius to obtain a length-scaled covering,

$$f_{\rm cov}^{\rm vol} \equiv R\,\langle\kappa_{\rm cov}\rangle_V = \frac{R}{V_{\rm box}}\int n_c(r)\,\pi\,r_c(r)^2\,dV = \frac{3\pi}{R^2}\int_0^R n_c(r)\,r_c(r)^2\,r^2\,dr. \tag{8}$$

Uniform limit. $f_{\rm cov}^{\rm vol,\,uniform} = R\,n_c^{(0)}\,\pi\,r_c^2 = \frac{3}{4}\,N\,(r_c/R)^2$, identical to Eq. (2).

Physical meaning. A bulk diagnostic of how much projected clump area exists in the box, normalized by the domain size. The radial weight r^2 heavily favors outer clumps over inner clumps and hence differs from the LOS integral whenever the profile is non-trivial.

4. Equivalence in the Uniform Limit

In the uniform-clump limit the three definitions are related by simple constants, summarized below:

Definition	Symbol	Uniform-limit value
Radial LOS (current LaRT)	$f_{\rm cov}^{\rm LOS}$	$\frac{3}{4}N(r_c/R)^2$
Impact-parameter average	$\langle f_{\rm cov}\rangle_b$	$N(r_c/R)^2$
Volume-weighted	$f_{\rm cov}^{\rm vol}$	$\frac{3}{4}N(r_c/R)^2$

The radial-LOS and volume-weighted definitions coincide in the uniform case but differ in general (the volume weight $4\pi r^2$ is absent from the LOS form). The impact-parameter average exceeds the LOS form by a constant factor $4/3$ in the uniform limit.

5. Behavior for Non-Uniform Profiles

The three averages diverge once r_c or n_c depend on position. Two illustrative cases:

(a) Thin shell of clumps at $r = r_0$. Let $n_c(r) = (N/4\pi r_0^2)\,\delta(r - r_0)$ with constant r_c.

- LOS: $f_{\rm cov}^{\rm LOS} = (N/4\pi r_0^2)\,\pi\,r_c^2 = N r_c^2/(4r_0^2)$.
  Independent of R; only the shell radius matters.

- Volume-weighted: $f_{\rm cov}^{\rm vol} = (3\pi/R^2)\,(N/4\pi r_0^2)\,r_c^2\,r_0^2$ (shell width factor).
  Suppressed by $(r_0/R)^0$ but multiplied by r^2 inside the integral, so a shell at large r_0 contributes more than a shell at small r_0.

- Impact-parameter average: depends on r_0/R through the chord-length geometry.

(b) Power-law clump density $n_c(r) \propto r^{-\alpha}$. With constant r_c and $\alpha < 3$ to keep the total finite,

$$f_{\rm cov}^{\rm LOS} \propto \int_0^R r^{-\alpha}\,dr \;\propto\; R^{1-\alpha}, \tag{9}$$

$$f_{\rm cov}^{\rm vol} \propto \int_0^R r^{-\alpha} r^2\,dr \;\propto\; R^{3-\alpha}. \tag{10}$$

The LOS integral is dominated by inner clumps (for $\alpha > 1$), while the volume-weighted form is dominated by outer clumps for any $\alpha < 3$. This contrast is the key reason the two definitions cannot be used interchangeably outside the uniform limit.

6. Recommendation for the Generalized Implementation

The radial line-of-sight integral, Eq. (5), is the recommended default because:

1. It coincides with the current LaRT formula Eq. (2) in the uniform limit, so existing input files retain their meaning.

2. It is the quantity most directly probed by a centrally located photon source; for radiative-transfer diagnostics this is a more faithful indicator than a volume-weighted bulk number.

3. It is a single one-dimensional integral and is straightforward to evaluate numerically in the helper routine `derive_clump_normalization`.

The impact-parameter average, Eq. (6), can optionally be reported alongside as a diagnostic for comparison with absorption-line observations of background sources. The volume-weighted definition, Eq. (8), is included here for completeness but is not recommended as the primary input quantity because of its strong outer-shell weighting.